

# Multilayer Perceptron (MLP) Implementation in Python

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import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score, confusion_matrix

# Load dataset
data = load_iris()
X = data.data
y = data.target

# Split dataset into training and testing
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=1)

# Create MLP model
mlp = MLPClassifier(hidden_layer_sizes=(5,4),
                    max_iter=1000,
                    learning_rate_init=0.01,
                    random_state=1)

# Train the model
mlp.fit(X_train, y_train)

# Make predictions
y_pred = mlp.predict(X_test)

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)

# Confusion matrix
cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:")
print(cm)
```